



Health Insurance Management

PROJECT REPORT

Health Insurance Management System

SPECIFICATION, PLANNING, ER DIAGRAM, TESTING, AND FINAL CHECKLIST

This report completes the missing Aptech-style documentation sections for the Health Insurance Management System using the implemented application workflow and the real Entity Framework database schema.



Students on This Project

STUDENT1623471

MUHAMMAD RUTAAB ALI

STUDENT1622741

ABDUL REHMAN KHAN

STUDENT1629514

MUHAMMAD BILAL

STUDENT1625350

SYED MUHAMMAD FARAZ



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Problem Definition

PROJECT PURPOSE

The Health Insurance Management System is an online application for an organization that wants to manage employee health insurance digitally. The system stores employee information, insurance company records, policy details, employee policy requests, bill approvals, payment processing, and status tracking in one role-based web application.

The earlier process depended on manual records, spreadsheets, scattered software, and repeated follow-up between employees, administrators, managers, and finance staff. This made the workflow time consuming, less secure, difficult to track, and hard to report accurately.

EXISTING ISSUES

- + Manual work creates delays and duplicate effort.
- + Employee and policy information is difficult to maintain centrally.
- + Request and bill status is not visible to each role in real time.
- + Security is weak without proper authentication and role-based access.

PROPOSED SOLUTION

- + Provide a secure login-based system for Admin, Employee, Manager, and Finance Manager.
- + Maintain employee, company, policy, request, approval, and bill data in a database.
- + Support policy search, request submission, approval, payment, and closure workflows.
- + Generate reports and maintain notification logs for important workflow events.

EXPECTED BENEFIT

- + Reduces manual work and repeated communication.
- + Improves data accuracy, security, and traceability.
- + Makes policy and bill status easier to monitor.
- + Gives each role a focused dashboard and workflow.



Customer Requirement Specification

ADMIN REQUIREMENTS

- + Register, update, activate, deactivate, and manage employees.
- + Add and maintain insurance company records.
- + Add, update, activate, deactivate, and manage policies.
- + Assign policies to employees and review policy requests.
- + Generate reports for employee, request, bill, and date-range activity.

EMPLOYEE REQUIREMENTS

- + Login securely using username and password.
- + View and update personal details.
- + Change password using a protected workflow.
- + Search available policies by requirement.
- + Submit policy requests with bill amount and reason.
- + Track request status from pending to closed or rejected.

MANAGER REQUIREMENTS

- + View employee insurance requests and generated bills.
- + Review request details before making a decision.
- + Approve or reject requests with remarks.
- + Forward approved bills to the Finance Manager.
- + Maintain profile information through the role dashboard.

FINANCE MANAGER REQUIREMENTS

- + Receive bills forwarded by managers.
- + Review bill amount, request details, and policy information.
- + Credit payment for accepted bills.
- + Close requests after payment is completed.
- + Maintain profile information through the role dashboard.

SYSTEM REQUIREMENTS

- + Authenticate users and derive role access on the server side.
- + Maintain status changes for policy requests, approvals, and bills.
- + Store password reset tokens securely and expire them after use or timeout.
- + Log notification events and delivery attempts.
- + Validate important form fields before saving data.

REPORTING REQUIREMENTS

- + Filter reports by date or date range.
- + Display employee, request, and bill records clearly.
- + Support administrative monitoring of workflow progress.
- + Keep output readable in responsive and printable layouts.



Project Plan

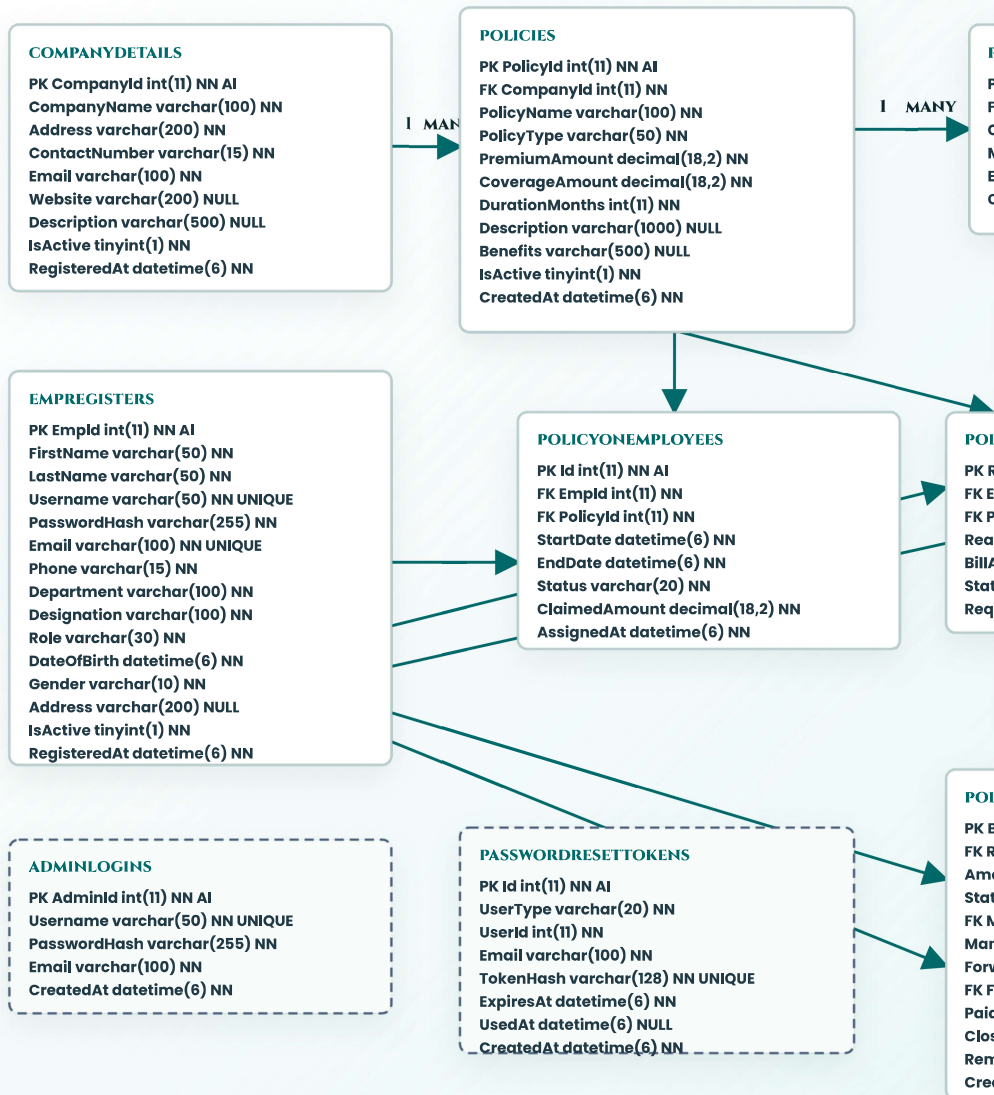
PHASE	ACTIVITIES	DELIVERABLES	REVIEW POINT
Requirement Analysis	Study problem statement, roles, inputs, outputs, and workflow expectations.	Requirement notes and module list.	Confirm Admin, Employee, Manager, and Finance flows.
Database Design	Define entities for users, companies, policies, assignments, requests, approvals, bills, and logs.	EF models and real ER diagram.	Check keys, foreign keys, and one-to-one workflow tables.
UI and Workflow Design	Design public pages, role dashboards, forms, tables, and status views.	User Guide and Design Guide patterns.	Confirm each role can reach expected tasks.
Implementation	Build controllers, views, validation, auth, notifications, and workflow actions.	ASP.NET Core MVC application.	Check CRUD, request, approval, payment, and report flows.

PHASE	ACTIVITIES	DELIVERABLES	REVIEW POINT
Testing	Perform static and manual checks for forms, role access, workflow states, and reports.	Unit Testing Checklist and Final Checklist.	Verify required modules are ready for submission.
Documentation	Prepare user, developer, design, and project report documents.	Project documentation folder.	Confirm Aptech required sections are present.



E-R Diagram

Schema Source: This diagram follows the supplied MySQL dump `healthinsurancemanagementdb.sql` . **Data rows are ignored; only tables, columns, keys, and foreign-key constraints are shown. NN means not null, AI means auto increment.**



RELATIONSHIP	CARDINALITY	IMPLEMENTATION DETAIL
policies.CompanyId to companydetails.Com panyId	Many to one	FK_Policies_CompanyDetai ls_CompanyId

Relationship	Cardinality	Implementation Detail
policyapprovaldetails. ManagerId to empregisters.EmpId	Many to one	FK_PolicyApprovalDetails_ _EmpRegisters_ManagerId
policyapprovaldetails. RequestId to policyrequestdetails.R equestId	One to one	Unique index on RequestId ; cascade delete.
policybills.FinanceMan agerId to empregisters.EmpId	Many to one	Nullable finance manager reference.
policybills.ManagerId to empregisters.EmpId	Many to one	Nullable manager decision reference.
policybills.RequestId to policyrequestdetails.R equestId	One to one	Unique index on RequestId ; cascade delete.
policyonemployees.E mpId to empregisters.EmpId	Many to one	Cascade delete from employee to assigned policies.
policyonemployees.Po licyId to policies.PolicyId	Many to one	Policy assignment reference.
policyrequestdetails.E mpId to empregisters.EmpId	Many to one	Cascade delete from employee to requests.
policyrequestdetails.P olicyId to policies.PolicyId	Many to one	Request references selected policy.

RELATIONSHIP	CARDINALITY	IMPLEMENTATION DETAIL
policytotaldescriptions .PolicyId to policies.PolicyId	Many to one	Cascade delete from policy to detailed descriptions.
adminlogins, hospitalinfos, passwordresettokens, emailnotificationlogs, __efmigrationshistory	No hard FK in dump	Support, standalone, or framework- maintained tables.



Algorithms

LOGIN AUTHENTICATION

1. Accept username and password.
2. Search admin and staff account stores.
3. Verify the stored password hash.
4. Create session values for user id, role, and display name.
5. Redirect to the role dashboard.

EMPLOYEE REGISTRATION

1. Admin enters employee profile and password.
2. Validate name, username, email, phone, role, and date of birth.
3. Hash password before saving.
4. Create employee record with active status.
5. Notify related staff if configured.

POLICY CREATION

1. Admin selects insurance company.
2. Enter policy name, type, premium, coverage, and duration.
3. Validate numeric and required fields.
4. Save policy linked to the company.
5. Show policy in search and management tables.

POLICY REQUEST

1. Employee searches available policies.
2. Employee opens details and submits request reason and bill amount.
3. System creates a pending request.
4. System creates or prepares bill workflow data.
5. Manager queue receives the request.

MANAGER APPROVAL

1. Manager opens pending request details.
2. Manager approves or rejects with remarks.
3. Approval record stores manager decision.
4. Approved request can be forwarded to finance.
5. Rejected request returns final rejected status.

FINANCE PAYMENT

1. Finance Manager reviews forwarded bills.
2. Finance validates amount and request context.
3. Finance credits payment.
4. System marks bill paid.
5. Finance closes the request after payment.

PASSWORD RESET

1. User enters registered email.
2. System identifies admin or staff account.
3. Create secure hashed reset token.
4. Email reset link to the registered recipient.
5. Validate token before allowing password update.

REPORT GENERATION

1. Admin selects report filters and date range.
2. System validates date inputs.
3. Query matching employees, requests, or bills.
4. Display results in readable tables.
5. Show empty state if no records match.



Task Sheet

MODULE	TASK	RESPONSIBLE ROLE	OUTPUT	STATUS
Authentication	Login, logout, forgot password, reset password.	System / User	Secure role session.	COMPLETED
Company Management	Add, edit, activate, deactivate, and delete companies.	Admin	Maintained company records.	COMPLETED
Policy Management	Add, edit, activate, deactivate, delete, and search policies.	Admin / Employee	Usable policy catalog.	COMPLETED
Employee Support	Register employees, update details, assign policies.	Admin / Employee	Employee profile and assignment records.	COMPLETED
Request Workflow	Create employee	Employee	Pending policy	COMPLETED

MODULE	TASK	RESPONSIBLE ROLE	OUTPUT	STATUS
	request and bill amount.		request.	
Manager Workflow	Approve, reject, and forward bills.	Manager	Decision and forwarded bill status.	COMPLETE D
Finance Workflow	Credit payment and close request.	Finance Manager	Paid and closed request.	COMPLETE D
Reports	Filter and display report data.	Admin	Date-range report tables.	COMPLETE D
Documentation	Prepare user, developer, design, and project report documents.	Project Team	Submission-ready guide files.	COMPLETE D



Project Review and Monitoring Report

REVIEW STAGE	REVIEW FOCUS	EVIDENCE	REMARKS
Review 1	Problem understanding, role identification, and module breakdown.	Problem definition and requirement specification.	Roles and modules confirmed from source specification.
Review 2	Database structure and workflow relationships.	EF Core models, DbSets, and ER diagram.	Implemented schema includes request, approval, bill, notification, and reset-token tables.
Review 3	Role dashboards, forms, tables, and workflow screens.	User Guide, Design Guide, and visible project pages.	UI follows the shared Material-style documentation theme.
Review 4	Testing readiness and final report completeness.	Unit Testing Checklist and Final Checklist.	Documentation includes all missing specification sections.



Unit Testing Checklist

AUTHENTICATION

- + Valid users can login.
- + Invalid passwords are rejected.
- + Role session redirects to correct dashboard.
- + Password reset accepts only valid tokens.

ADMIN CRUD

- + Company create/update/delete works.
- + Policy create/update/delete works.
- + Employee registration validates required fields.
- + Hard delete blockers show safe messages.

EMPLOYEE FLOW

- + Employee can view and update details.
- + Policy search filters return matching records.
- + Policy details show company and coverage data.
- + Insurance request stores bill amount and reason.

MANAGER FLOW

- + Manager can view pending requests.
- + Approval creates decision record.
- + Rejection stores remarks.
- + Approved bills can be forwarded to finance.

FINANCE FLOW

- + Finance receives forwarded bills.
- + Credit action records payment state.
- + Close action marks request complete.
- + Profile pages remain role protected.

REPORTS AND LOGS

- + Date filters validate input range.
- + Report tables show matched records.
- + Empty reports show a readable empty state.
- + Email notification attempts are logged.



Final Checklist

CHECKLIST ITEM	EXPECTED EVIDENCE	STATUS
Certificate of Completion	Developer Guide certificate section.	READY
Table of Contents	Each guide has a navigation section.	READY
Problem Definition	This report documents the system problem and proposed solution.	READY
Customer Requirement Specification	Actor-based requirement section in this report.	READY
Project Plan	Phase table from analysis to documentation.	READY
E-R Diagram	Real EF Core schema diagram included in this report.	READY
Algorithms	Core workflow algorithms included.	READY
GUI Standards Document	Design Guide documents typography, palette, layout, tables, and forms.	READY

CHECKLIST ITEM	EXPECTED EVIDENCE	STATUS
Interface Design Document	User Guide and Design Guide document public and role interface behavior.	READY
Task Sheet	Task table included in this report.	READY
Project Review and Monitoring Report	Milestone review table included in this report.	READY
Unit Testing Checklist	Checklist section included in this report.	READY
Final Check List	This section confirms submission readiness.	READY